PRESIDENCY UNIVERSITY

GAIN MORE KNOWLEDGE REACH GREATER HEIGHTS

Odd Semester 2025-2026

School of Engineering

Syllabus Announcement of Mid-Term Examination-FEB- 2026

Course Title: Principles of Quantum Mechanics and quantum computing **Course Code**: PHY 3001

Instructor In-Charge: Dr. G Srinivas Reddy Date of Announcement: 9th FEB 26

Exam Date: 12-03-2026, **Max. Marks: 50 Weightage: 25% Time: 9:30-11:00 Am**

It is hereby notified that, the mid-term examination for Principles of Quantum Mechanics and quantum computing (PHY3001 EVEN SEM) of AY 2025-26 will be held on 1**2th February 2026.**

Syllabus:

Module 1: Fundamentals of Quantum Mechanics: Introduction to quantum mechanics and historical development, Dual nature of radiation and matter, de Broglie hypothesis, Wave function, Schrödinger's time-independent wave equation, Hilbert space: vectors, states, and basic functions, Probability density and normalization of wave functions, Dirac Ket–Bra notation, Operators in quantum mechanics, Pauli matrices, Conjugate , transpose, and Hermitian operators, Unitary matrices. orthogonality, probability calculation, Inner product, outer product.

Module 2: Principles of Quantum Computing and Qubit Operations

Introduction to quantum computing, Moore's law & its end, Differences between classical and quantum computing, Concept of Qubit and its properties, Representation of qubit by Bloch sphere, Quantum Gates: Single Qubit Gates: Quantum Not Gate, Pauli Z Gate, Hadamard Gate, Phase Gate (or S Gate), T Gate. Multiple Qubit Gates: Controlled gate - CNOT Gate, (Discussion for 4 different input states). Representation of Swap gate, Controlled - Z gate, Toffoli gate. FREDKIN Gate.

Note:

The pattern of examination:

Part A: Knowledge and thought-provoking (5Q × 2M=10M)

Part B: Understanding and applying (4Q×10M=40M)

Dr G Srinivas Reddy

Signature of Instructor In-Charge